REVISION GROUP WORK: TWO-MEMBER GROUPS

Students are required to form **groups of two members** to conduct revision group work. Each group should collaboratively review the relevant course materials, discuss key concepts, and prepare comprehensive answers to the revision questions.

Submission Deadline: 16:30

Please ensure that your group work is completed, reviewed, and submitted **before 16:30**.

QUESTION 1: Digital Education and AI Tutoring

A rural secondary-school network intends to introduce an AI-powered mobile tutoring platform to improve mathematics performance among students with limited access to qualified teachers.

- **a.** Critically formulate a research problem for investigating the effectiveness of the AI tutoring platform. Explain the characteristics that make the problem researchable, feasible, significant, and contextually relevant. **(5 marks)**
- **b.** Develop three research objectives and two research questions aligned with the proposed study. **(5 marks)**
- **c.** Formulate appropriate null and alternative hypotheses for examining the relationship between AI tutoring and students' academic performance. **(5 marks)**
- **d.** Justify whether a quantitative, qualitative, or mixed-methods research approach would be most appropriate. **(5 marks)**
- **e.** Discuss the ethical issues associated with collecting academic and behavioral data from school students, including consent, privacy, confidentiality, and data protection. **(5 marks)**
- **f.** Design context diagram and activity diagram for digital education using AI **(5 marks)**

QUESTION 2: Smart Traffic Management in Kigali

The City of Kigali plans to deploy an intelligent traffic management system integrating GPS information, traffic cameras, IoT sensors, and intelligent traffic lights to reduce congestion and improve road safety.

- **a.** Formulate a clear research problem and explain how it could be narrowed into a feasible academic investigation. **(5 marks)**
- **b.** Identify suitable primary and secondary data sources for the study and distinguish between them using relevant examples. **(5 marks)**
- **c.** Design an appropriate sampling strategy for selecting roads, vehicles, drivers, or commuters for the investigation. Justify your choice. **(5 marks)**
- **d.** Explain how correlation, regression, or classification techniques could be applied to investigate traffic congestion and predict traffic conditions. **(5 marks)**

e. Critically discuss ethical concerns related to GPS tracking, surveillance cameras, vehicle identification, and commuter data. **(5 marks)**
f. Design Dataflow diagram, and class diagram for smart traffic management **(5 marks)**

MODEL QUESTION 3: AI-Based Smart Agriculture

A research team proposes an AI system that uses rainfall, temperature, soil characteristics, satellite imagery, and historical agricultural records to predict crop yields for smallholder farmers in Rwanda.

a. Develop a suitable research aim and four measurable research objectives for the proposed investigation. **(5 marks)**
b. Formulate a testable research hypothesis concerning the effect of AI-based forecasting on crop-yield prediction accuracy. **(5 marks)**
c. Explain the difference between **inductive and deductive reasoning** and demonstrate how each could be applied in this research. **(5 marks)**
d. Justify the use of stratified sampling based on agro-ecological zones, crop types, or soil characteristics. **(5 marks)**
e. Explain how regression analysis, correlation analysis, or principal component analysis could identify important predictors of crop productivity. **(5 marks)**
f. Design flowchart, and collaration diagram for AI based smart agriculture **(5 marks)**

QUESTION 4: Cybersecurity Awareness in SMEs

A group of SMEs has experienced increasing phishing attacks and unauthorized access incidents. Management proposes an employee cybersecurity-awareness programme and wishes to evaluate whether the intervention improves security behavior.

a. Explain how a literature review could be used to identify the research gap and establish the theoretical foundation for the study. **(5 marks)**
b. Develop a research framework showing the relationship between cybersecurity training, employee knowledge, security behavior, and security incidents. **(5 marks)**
c. Formulate appropriate H_0 and H_1 hypotheses for evaluating the effectiveness of the training programme. **(5 marks)**
d. Explain when a Chi-square test, t-test, or F-test/ANOVA would be appropriate for analyzing the study data. **(5 marks)**
e. Discuss ethical challenges associated with collecting employee cybersecurity-performance data and propose appropriate safeguards. **(5 marks)**
f. Design and describe flowchart, and entity relationship diagram for cybersecurity awareness **(5 marks)**

QUESTION 5: Digital Health and Community Health Workers

A health organization intends to introduce a mobile health application through which community health workers can record patient information, monitor treatment adherence, and submit disease-surveillance reports.

- **a.** Compare cross-sectional and longitudinal research designs and determine which would be more appropriate for evaluating the intervention over time. **(5 marks)**
- **b.** Identify suitable primary and secondary data sources and recommend appropriate instruments for collecting the required information. **(5 marks)**
- **c.** Explain how data cleaning, missing-value treatment, normalization, and validation should be performed before statistical analysis. **(5 marks)**
- **d.** Explain how correlation and regression analysis could be used to determine whether application usage predicts patient adherence. **(5 marks)**
- **e.** Develop a comprehensive ethical framework covering informed consent, confidentiality, anonymization, secure storage, and responsible use of patient information. **(5 marks)**

QUESTION 6:Blockchain-Based Land Registration

A government agency is considering replacing its paper-based land-registration system with a blockchain-based platform designed to reduce fraud, improve transparency, and shorten registration time.

- **a.** Critically formulate a research problem addressing inefficiencies in the existing land-registration system. **(5 marks)**
- **b.** Develop appropriate research objectives and research questions for investigating blockchain adoption in land administration. **(5 marks)**
- **c.** Explain the sequential process of hypothesis testing, from formulation of H_0 and H_1 to statistical decision-making. **(5 marks)**
- **d.** Compare appropriate parametric and non-parametric tests that could be applied to blockchain-adoption data. **(5 marks)**
- **e.** Discuss legal, ethical, intellectual-property, data-ownership, and jurisdictional issues associated with blockchain-based land records. **(5 marks)**

QUESTION 7: Multigranularity Deep Learning for LULC Classification

A research team proposes a **multigranularity deep-learning model** for classifying Land Use and Land Cover (LULC) from satellite imagery in Kigali. The model is intended to integrate fine-, medium-, and coarse-scale spatial features to distinguish built-up areas, agriculture, forests, wetlands, water bodies, and bare land.

- **a.** Formulate a research problem that identifies the limitations of conventional single-scale LULC classification approaches. **(5 marks)**
- **b.** Develop three research objectives and two research questions for evaluating the proposed multigranularity model. **(5 marks)**

- **c.** Identify the independent, dependent, control, and possible confounding variables in the proposed research. **(5 marks)**
- **d.** Formulate appropriate null and alternative hypotheses for comparing the proposed model with a conventional deep-learning classifier. **(5 marks)**
- **e.** Explain appropriate performance measures, such as accuracy, precision, recall, F1-score, Intersection over Union, and confusion matrices, for evaluating the classification model. **(5 marks)**

QUESTION 8: Smart City Waste Management

Kigali intends to develop an intelligent waste-management system using IoT-enabled waste bins, GPS-enabled collection vehicles, mobile applications, and predictive analytics to optimize waste collection.

- **a.** Critically identify and justify a research problem related to inefficiencies in conventional waste-management operations. **(5 marks)**
- **b.** Formulate a research aim, four objectives, and two research questions for the proposed investigation. **(5 marks)**
- **c.** Identify suitable primary and secondary data sources and recommend appropriate data-collection instruments. **(5 marks)**
- **d.** Explain how regression or machine-learning classification could be used to predict waste-generation patterns or optimize collection schedules. **(5 marks)**
- **e.** Evaluate the ethical and practical challenges associated with collecting location, household, and municipal-service data. **(5 marks)**

QUESTION 9: E-Learning and Student Performance

A university has introduced an LMS that records login frequency, time spent on learning materials, assignment submissions, discussion participation, and examination results. Researchers want to determine whether LMS engagement predicts academic performance.

- **a.** Explain how a literature review could be used to identify the research gap and develop a conceptual framework for the study. **(5 marks)**
- **b.** Identify the **independent, dependent, mediating, moderating, and control variables** that could be incorporated into the research model. **(5 marks)**
- **c.** Formulate appropriate directional and non-directional hypotheses for investigating LMS engagement and academic performance. **(5 marks)**
- **d.** Explain how correlation and multiple regression could be used to examine the relationships among the variables. **(5 marks)**
- **e.** Discuss possible sources of bias and confounding variables, and explain how the research design could minimize their influence. **(5 marks)**

QUESTION 10: AI-Based Healthcare Diagnosis

A hospital proposes an AI-based diagnostic system that analyzes patient symptoms, laboratory results, medical images, and clinical records to assist healthcare professionals in identifying diseases at an early stage.

- **a.** Critically formulate a research problem that addresses the need for AI-assisted diagnosis while identifying a specific research gap. **(5 marks)**
- **b.** Develop appropriate research objectives and research questions for evaluating the diagnostic system. **(5 marks)**
- **c.** Explain the difference between **quantitative, qualitative, and mixed-methods research**, and justify the most appropriate approach for this investigation. **(5 marks)**
- **d.** Explain how sampling, data preprocessing, hypothesis testing, and model evaluation should be incorporated into the research design. **(5 marks)**
- **e.** Analyze the ethical implications of using patient records and AI-generated diagnostic recommendations, including informed consent, privacy, bias, explainability, accountability, and potential harm. **(5 marks)**